

18	A	For destructive interference to first occur, the path difference of the two waves must be half of a wavelength. When the sliding tube is raised, the difference between paths 1 and 2 is the additional vertical sections on both sides of the tube above. Hence, the path difference is $2(a_1 - b)$ where a_1 is the distance a when the first minimum is detected. Hence, $0.5\lambda = 2(a_1 - b)$ and $a_1 = 0.25\lambda + b$. Since the initial distance of a is equal to b, a has increased by 0.25λ.